

# Daniel Cher

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## Education

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**Washington University in St. Louis**

*Ph.D. in Computational Data Sciences*

St. Louis, MO

Sep 2024 – present

**Worcester Polytechnic Institute**

*M.S. in Data Science*

Worcester, MA

Jan 2019 – Dec 2021

**Boston University**

*B.A. in Economics*

Boston, MA

Sep 2013 – May 2017

*B.S. in Business Administration, Concentration in Finance and Accounting*

## Honors & Fellowships

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**Best Paper Award**, TerraBytes II Workshop, ECCV

2026

**AI-ACCESS Fellow**, Washington University in St. Louis

2024 – present

**Olin Chancellor's Fellow**, Washington University in St. Louis

2024 – present

## Publications

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### Peer-Reviewed

- [1] **Cher, D.**, Iqbal, H., Xing, E., Wei, B., Jacobs, N. *Tessellating the Earth: Learnable Spherical Voronoi Partitions for Location Encoding*. **ECCV'26**. 📄 Paper 📄 arXiv:2606.27514 🔗
- [2] Wei, B.<sup>\*</sup>, Sastry, S.<sup>\*</sup>, **Cher, D.**<sup>\*</sup>, Xing, E., Jacobs, N. *TerraDiT-Ω: Unified Spatial Control for Satellite Image Synthesis with Any Geospatial Primitive*. **ECCV'26**. 📄 Paper 📄 arXiv:2606.31029 🔗
- [3] Sastry, S., Khanal, S., Dhakal, A., Lin, J., **Cher, D.**, Jarosz, P., Jacobs, N. *ProM3E: Probabilistic Masked MultiModal Embedding Model for Ecology*. **CVPR'26**. 📄 arXiv:2511.02946 🔗
- [4] **Cher, D.**<sup>\*</sup>, Wei, B.<sup>\*</sup>, Sastry, S., Jacobs, N. *VectorSynth: Fine-Grained Satellite Image Synthesis with Structured Semantics*. **WACV'26**, Tucson, AZ. 📄 arXiv:2511.07744 🔗
- [5] Khanal, S., Cui, Y., **Cher, D.**, Xing, E., Wei, B., Sastry, S., Jacobs, N. *Genesis: A Generative Engine for Hierarchical Satellite Image Synthesis*. **SIGSPATIAL'26, Oral Presentation**. 📄 arXiv:2609.02683
- [6] Sastry, S.<sup>\*</sup>, **Cher, D.**<sup>\*</sup>, Wei, B.<sup>\*</sup>, Dhakal, A., Khanal, S., Gupta, D., Jacobs, N. *TerraDiT: Point-Conditioned Diffusion Transformer for Satellite Image Synthesis*. **ECCVW'26 (TerraBytes II), Best Paper Award, Oral Presentation**. 📄 arXiv:2603.02172 🔗

### Preprints & Under Review

- [7] **Cher, D.**, Xing, E., Li, K., Wei, B., Corley, I., Jacobs, N. *Chronosphere: Space-Time Tessellation of Local Climate Experts*. **Under Review**. 📄 arXiv:2609.21872
- [8] **Cher, D.**, Wei, B., Xing, E., Jarosz, P., Sandvig, R., Jacobs, N. *PhenoMap: Phenological Neural Fields for Ecological Mapping*. **Under Review**.
- [9] Sastry, S.<sup>\*</sup>, **Cher, D.**<sup>\*</sup>, Wei, B., Khanal, S., Gao, T., Dhakal, A., Xing, E., Jacobs, N. *TaxaWalk: Species Recognition as a Walk Down the Tree of Life*. **Under Review**.
- [10] Xing, E., Li, V., **Cher, D.**, Jacobs, N. *QuASI: Query-Adaptive Sparse Indexing for Search*. **Under Review**.
- [11] Wei, B., Hsu, C.-C., Gao, T., **Cher, D.**, Xing, E., Beaulieu, D., Jacobs, N. *X Marks the Spot: Diffusion for Image Geolocalization with Location Representations*. **Under Review**.
- [12] Xing, E., Wei, B., **Cher, D.**, Jacobs, N. *Learning to Allocate Tokens for Vision Transformers*. **Under Review**.

\* denotes equal contribution.

## Presentations & Technical Reports

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- [1] **Cher, D.**, Kolouju, P., Xing, E., Pless, R., Stylianou, A., Jacobs, N. 2026. *Efficient Hierarchical Search for Scalable Retrieval in Global Remote Sensing Imagery*. **Oral Presentation** at IGARSS 2026.
- [2] **Cher, D.** 2023. *Geographic Rating in Insurance Pricing*. White paper written for internal product pricing modeling groups. Disseminated companywide to local pricing modeling teams.
- [3] **Cher, D.**, Meng, S. 2022. *Leverage Unlabeled Data with Contrastive Learning*. Session presented at the 11th Annual Liberty Mutual Data Science Forum, Boston, MA.
- [4] **Cher, D.** 2021. *Advances in Geospatial Residual Modeling: Leveraging luigi for Experimentation*. Session presented at the 10th Annual Liberty Mutual Data Science Forum, Boston, MA.
- [5] **Cher, D.**, Dichiaro, A., Alsaedi, M., Tang, S. 2021. *Modeling PFAS risk predictions and source attribution for private drinking water wells in MA: A multi-faceted study on toxic forever chemicals using computational analysis and machine learning*. Study disseminated to MassDEP stakeholders. Project presented at the 5th Annual Graduate Research Innovation Exchange, Worcester, MA. 

## Working Experience

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### Washington University in St. Louis

St. Louis, MO

*Ph.D. Student, Multimodal Vision Research Laboratory*

Sep 2024 – present

- Developed a location encoder using learnable spherical Voronoi partitions with global semantic tokens, contrastively aligned to satellite imagery. Set state of the art across a geospatial benchmark suite. Released code and pretrained models on Hugging Face. (ECCV'26)
- Built diffusion transformer models that synthesize satellite imagery conditioned on arbitrary geospatial primitives (points, boxes, polylines, polygons) through geometry-aware local attention. Synthetic data improved downstream land-cover segmentation, object detection, and road extraction. (ECCV'26, WACV'26)
- Mentored four undergraduate researchers (three through the Center for the Environment Summer Undergraduate Research Program, one through a WashU-funded REU) and one M.S. student.

### Liberty Mutual

Boston, MA

*Assistant Director, Data Science — Geospatial Modeling*

Jan 2022 – Apr 2024

- Improved geographic price segmentation by incorporating geodemographic similarity into proprietary smoothing algorithms and engineering features from open-source raster data (weather, terrain, land cover). Packaged algorithms as scikit-learn regressors to standardize the modeling interface and enable Optuna hyperparameter search. 2.5–4% relative loss ratio improvement on US auto and home businesses (\$1B+ book of business).
- Managed four data scientists and led the *Geospatial Modeling* training series (ten lectures covering data sourcing through deployment) for ten data scientists across three international regions. Directed five geospatial pricing projects as manager and reviewer; each region then owned its own geospatial modeling.
- Authored *Geographic Rating in Insurance Pricing*, an internal white paper on quantifying geographic risk (feature engineering, variable selection, peril-specific modeling, clustering), distributed companywide.

*Data Scientist, Geospatial Modeling*

Sep 2017 – Jan 2022

- Lead developer of the territory-pricing pipeline (luigi + mlflow): ingests data, fits and compares territory-risk models, persists hyperparameters and artifacts to S3, and surfaces results in a shareable dashboard. Cut FTE modeling time 80% and made runs reproducible.
- Subject-matter expert for international territory pricing across Portugal, Colombia, Brazil, Spain, and Thailand. Performed spatial clustering, feature engineering from external datasets, and smoothing to correct spatial autocorrelation. 4% loss ratio improvement on a \$500M+ book of business in Brazil.

- Lead developer of **terripy**, the internal geospatial modeling package, which included geospatial data classes, feature engineering transformations, and proprietary algorithms, adopted companywide.
- Built an at-fault classifier over unstructured claim narratives with fast.ai, combining text and structured features. Developed with two teammates during a companywide data science competition and subsequently adopted by the business.

## Additional Research Experience

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### Aerial Imagery Research @ Liberty Mutual

Spring 2022

*Research project exploring benefits of contrastive learning on property roof computer vision models.*

- Researched contrastive learning architectures to decrease labeling needs of computer vision models. Found that contrastively trained model using domain-specific images outperformed general supervised pre-trained models like ones built on ImageNet.
- Developed contrastive learning models using clustering (SwAV), contrastive-instance (MoCoV2), and contrastive-network (BYOL) based architectures using PyTorch with PyTorch Lightning integrations.
- Tested pre-trained models using transfer learning to improve classification models of roof features including existence of solar panels and roof condition. Showed improved model performance by 2% F1 score.
- Presented research at the annual Data Science Forum in collaboration with the project sponsor.

## Skills

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**Methods:** diffusion transformers, contrastive and self-supervised learning, multimodal representation learning, vision-language models, location encoding, spatiotemporal modeling, remote sensing, machine learning, deep learning

**Coding Languages:** Python, R, SQL

**Libraries:** PyTorch, Lightning, Hugging Face (diffusers, transformers), open\_clip, xformers, torchgeo, geopandas, rasterio, GDAL, scikit-learn

**Tooling:** hydra, wandb, mlflow, Optuna, luigi, SLURM, Docker, AWS, Snowflake, Earth Engine, QGIS, ArcGIS, Git

**Spoken Languages:** Russian (fluent), Japanese (conversational)